The Agent Bazaar: An Empirical Test Bench for Virtual Economies

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December 12, 2025

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Submitted in partial fulfillment
of the requirements for the degree of

Bachelor of Science in Engineering

Department of Electrical and Computer Engineering
Princeton University

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Cameron Crow

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Abstract

As AI systems begin to participate directly in economic activity, understanding how LLM-powered agents behave in market settings has become increasingly important. Agent Bazaar introduces a simulation framework for studying fully agentic markets in which agents interact at high speed. Through scenarios such as price wars, first-mover advantages, and discriminatory pricing, the framework reveals both rational strategies and destabilizing behaviors that emerge when agents operate with limited information. Preliminary experiments show that firms can adopt competitive pricing tactics but may also fall into destructive strategies that reduce welfare in monopolistic environments. These early results highlight the potential for agentic markets to generate inequitable or harmful outcomes when guardrails are absent. Agent Bazaar provides a test bench for exploring these risks and evaluating interventions that may inform future regulation of agent-driven economic systems.

ACKNOWLEDGMENTS

I would like to thank Professor Chi Jin for his excitement and encouragement when I shared my ideas and asked if he would advise this senior thesis. Also special thanks to Seth Karten who has consistently held me accountable on this project and is ensuring that I and the project reach high standards.

Thank you to the Princeton Electrical and Computer Engineering Department and the School of Engineering and the Applied Sciences for providing funding for my undergraduate senior thesis.

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Abstract

As AI systems begin to participate directly in economic activity, understanding how LLM-powered agents behave in market settings has become increasingly important. Agent Bazaar introduces a simulation framework for studying fully agentic markets in which agents interact at high speed. Through scenarios such as price wars, first-mover advantages, and discriminatory pricing, the framework reveals both rational strategies and destabilizing behaviors that emerge when agents operate with limited information. Preliminary experiments show that firms can adopt competitive pricing tactics but may also fall into destructive strategies that reduce welfare in monopolistic environments. These early results highlight the potential for agentic markets to generate inequitable or harmful outcomes when guardrails are absent. Agent Bazaar provides a test bench for exploring these risks and evaluating interventions that may inform future regulation of agent-driven economic systems.

1 Introduction

On May 6, 2010, U.S. financial markets dropped 5–6% in a matter of minutes due to an unbound "sell algorithm." Amidst a period of economic worry and a morning of high volatility, a sell algorithm designed to execute actions at a rate of 9% the current market volume initiated a massive sell-off of roughly $4.1 billion worth of contracts. Known as the "Flash Crash", this sell-off was completed in 20 minutes, sending the market into a frenzy U.S. Securities and Exchange Commission & Commodity Futures Trading Commission (2010). Though this was limited to a high-frequency trading environment, there have been recent introductions of AI in retail. OpenAI, in collaboration with Stripe, announced Instant Checkout, where customers can use ChatGPT to discover products and complete transactions Weinstein & Kaliski (2025). Despite concerning history like the Flash Crash and the widespread integration of AI and LLMs into daily life—and, specifically, public retail—there is a lack of robust test benches for studying how agentic economies would behave. Without trialing of agentic markets, there remains a lack of empirical knowledge, thus risking a lack of preventative regulation surrounding their behavior and dangerous edge cases. Agentic economies potentially enable great inequality, favoring parties with access to advanced models, hardware, and other resources. In certain scenarios, such resources could allow for malicious market manipulation. Agent Bazaar introduces a simulation framework that enables experimentation with economic scenarios in fully agentic markets that can be utilized for research and regulation development. The framework emphasizes believable human-like behavior by the agents as both rational firms and consumers.

2 Related Work

Several recent projects explore the use of large language models as economic agents, but none provide a full simulation framework for testing economic scenarios with agent-driven demand.

VendingBench offers one of the earliest benchmark environments for evaluating whether an LLM can operate a simple business. The entrepreneurial agent manages money, purchases inputs, and sells products—capabilities that directly inspired the design of our FirmAgent. However, VendingBench focuses on isolated business tasks such as sending emails to suppliers rather than market-level behavior (Backlund & Petersson, 2025). In contrast, Agent Bazaar aims to simplify and extend VendingBench into a mutli-agent eBay-esque environment.

EconAgent moves toward richer macroeconomic behavior by equipping LLM agents with human-like traits and a perception module. This addition improves the agent's ability to interpret and respond to its environment (Li et al., 2024). We plan to imitate aspects of this module to strengthen our agents' reasoning and responsiveness.

Agent Bazaar has grown directly out of the LLM Economist project, which uses LLM-powered agents as workers who choose their labor allocation based on the current tax plan and their income, as well as a tax planner whose goal is to optimize the tax plan in order to maximize social welfare (Karten et al., 2025). Agent Bazaar aims to work in parallel with the LLM Economist by adding direct agent-agent interactions in a goods-based market.

These projects provide great contributions to the subject, but we can not find an empirical test bench for studying an agentic market system where both firm behavior and consumer demand are agent-generated. Existing research raises concerns about agentic economies, but lacks a platform for experimenting with price wars, inequality, convergence, or regulation. Agent Bazaar directly targets this gap by offering a simulation framework capable of testing economic scenarios before such systems appear in real-world markets.

3 Problem Setup

Agent Bazaar's core objective is to offer a simulation that allows effective evaluation of the emergent behaviors that arise in a fully agentic market in which LLM-powered agents are buying and selling amongst one another at high speeds. Such speed makes it difficult for human monitoring and correction to be effective. In these scenarios, the "best" agent can profit hugely and potentially manipulate the market by exploiting price wars, first-mover, and discriminatory pricing dynamics. Without empirical test benches, we cannot safely observe the movement of equity or surplus, or test the convergence (or lack thereof) of agentic economies. We examine three scenarios in particular.

3.1 Scenarios

3.1.1 Race to the Bottom

In the "Race to the Bottom" scenario, we expect price wars between firms. A seller agent can eliminate competition by dropping prices, potentially achieving a monopolistic state and exploiting consumers. Firms "open shop" for the day with a set of prices and inventory, fixing prices for the day. Consumers, when making orders, will find the best deals among all or a randomized subset of firms, submitting an order for the best one, specifying the firm they'd like to purchase from. The order is not filled upon submission. At the end of the day, orders are filled. This leads to all consumers attempting to purchase from the firm with the lowest price, driving prices down until prices are exactly equal to the marginal cost of production. Marginal cost is defined in Equation 1, where c is the input cost to produce that good, F is the operational cost of the firm, and q_i is the quantity produced. At this boundary, a firm is better off exiting the market with the money still in its account, and thus we assume that if the price dips below this equilibrium price, most firms will exit the market and this

cycle will continue until a single firm is still in operation.

$$p_t = c_t + F/Q_t \tag{1}$$

3.1.2 Early Bird

In the "Early Bird" or first-mover scenario, all of the surplus goes to the fastest consumers. The firms open shop at the start of the day. Consumers poll the firms, submitting an order for the best deal they find. The order is immediately filled upon submission (given it can be supplied and is a valid order). This scenario favors first–movers—the first consumers to place orders take all of the available surplus. In this setup, we expect to see a negative correlation between time of arrival/time of order and surplus received, both of which we can quantify and log. Surplus can be defined in Equation 2, where w_{it} is consumer i's willingness to pay (WTP) at timestep t.

$$S_{it} = w_{it} + p_{it} \tag{2}$$

3.1.3 Price Discrimination

In the price discrimination scenario, all of the surplus goes to the sellers. Each consumer's WTP is submitted as the price at which they'd like to make a purchase. Orders are filled by the firm with the price closest to but not exceeding the consumer's WTP, as indicated by the order submitted. In a market with many firms with prices distributed evenly among the lowest WTP and the highest WTP, this means the consumers receive little to no surplus, while the seller's profit is the difference between the customer's WTP and the cost of production.

These scenarios lead to massive inequality in the market, favoring parties with access to advanced models and resources, allowing for unfair competition. Understanding the dynamics that lead to such scenarios is essential in developing preventative regulation before agentic economies are fully deployed with bearing on the net worth of real people.

4 Methodology

In order to understand the emergent behaviors and dynamics of fully agentic markets, we build a framework for a simulation in which LLM-powered agents operate as firms, consumers, and sometimes both, buying and selling between each other. The current implementation expects that unregulated, high-speed agentic trading will regularly give rise to price wars, first-mover advantages, and surplus inequality. Our methodology isolates these effects by separating them into market setups that are more susceptible to such cases.

Our primary assumptions are that (i) an agent's primary goal is to maximize their own utility/profits as consumers/firms each time step and (ii) agents do not know other agents' internal states. These assumptions simplify pricing and consumption behavior.

4.1 Implementation

The current version of the simulation separates firms and consumers into FirmAgents and ConsumerAgents. ConsumerAgent actions (currently) are not LLM-powered like FirmAgent actions. ConsumerAgents' demand is instead derived from the cost of living (P_t) under constant elasticity of substitution (CES) homothetic preferences, where p_{gt} is the price of good g from the set of goods G at time t, α represents the consumers preference for good g at time t, and σ is the elasticity of substitution (Redding & Weinstein, 2020). By using CES taste parameters, we have a model that allows for heterogeneity among the ConsumerAgents even without prompting an LLM.

$$P_t = \left[\sum_{g \in G} \left(\frac{p_{gt}}{\alpha_{gt}} \right)^{1-\sigma} \right]^{\frac{1}{1-\sigma}}, \qquad \sigma > 1, \tag{3}$$

ConsumerAgents

Each ConsumerAgent possesses a natural language description of its persona. This description is used in a prompt to an LLM to generate a vector of taste parameters used for this consumer's CES utility. Each agent receives an income at the start of each timestep and pays taxes proportional to that income.

FirmAgents

FirmAgents are entirely LLM-powered with an LLM being prompted each time an action is taken. The action space consists of purchasing supplies, producing goods from those supplies, and setting prices for those goods. FirmAgents are also subject to taxes and an overhead cost for staying in business.

4.2 Simulation Loop

The simulation loop is visualized in detail in Figure 1

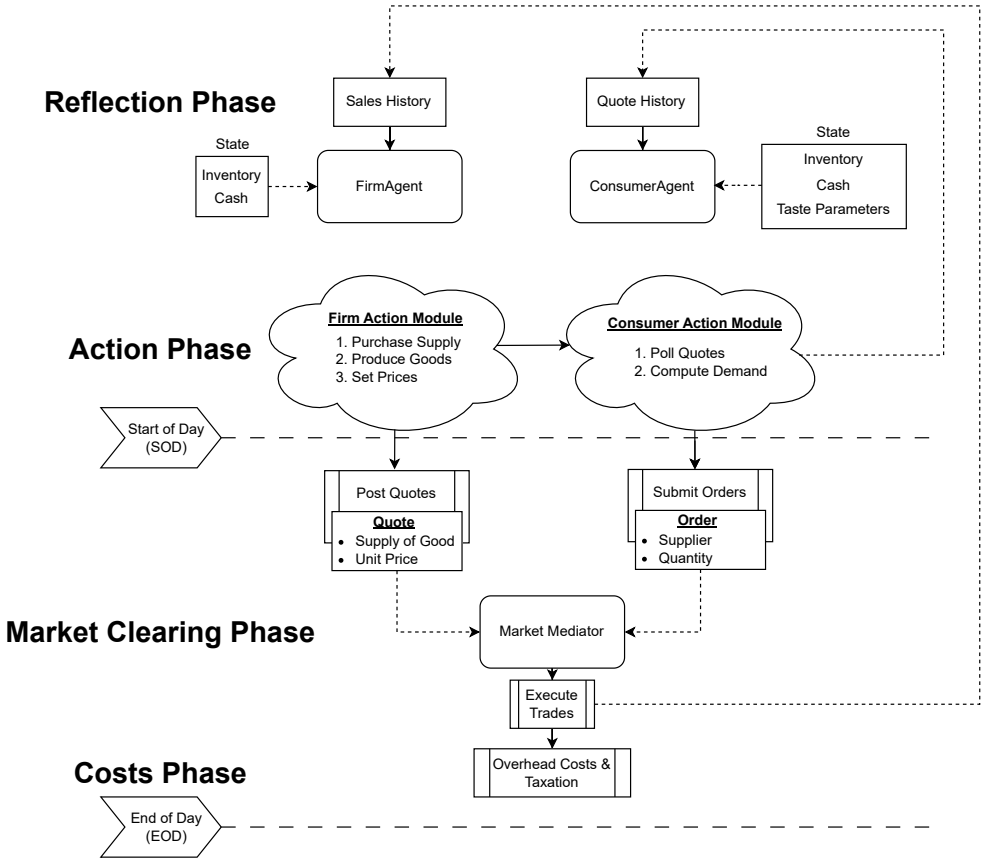

Figure 1: Agent Bazaar simulation control and information flow.

1. Firms purchase an amount of an aggregate supply.
2. Firms produce goods with their available supply.
3. Firms set the prices of the goods and post quotes to the market (sell contracts).
4. Consumers receive their income.

5. Consumers submit buy orders to the market by computing their cost of living and quantity demanded.
6. A mediator in the market matches orders and quotes, transferring cash and goods.
7. Taxation and overhead costs are applied.

4.3 Order Generation

Given prices and its taste parameters, each ConsumerAgent generates a quantity demanded for each good in that timestep using equation (x). The scenario determines the algorithm used to construct orders based on the quantity demanded.

4.4 Scenarios

To examine emergent behaviors, we simulate the three scenarios outlined in Section 3 (Race to the Bottom, Early Bird, and Price Discrimination) that can arise in a high-frequency trading agentic market in the absence of regulations. In these scenarios, the "best" agent can profit hugely and potentially manipulate the market. By experimenting with these scenarios, we will be able to (i) assess the validity of the simulation, (ii) assess the assumptions of these scenarios, and (iii) theorize about solutions to regulate agents in order to prevent exploitation.

4.5 Purpose

Across all experiments and trials, we are not searching for ways to optimize agent performance in the market by, say, optimizing the reflection module to yield the most profit. We are observing whether or not dangerous economic outcomes emerge. In doing so, we aim to assess possible regulations for real-world applications.

5 Preliminary Results

To gather initial results for the simulation, we ran a batch of experiements in a two-firm, N-consumer environment, with LLM-powered FirmAgents and CES-based ConsuemrAgents. These experiments are desinged to test our Race to the Bottom hypothesis: delayed market clearing and unregulated price competition will drive firms to undercut one another until a single firm monopolizes the market, leading to overall lower social welfare.

Trial 4

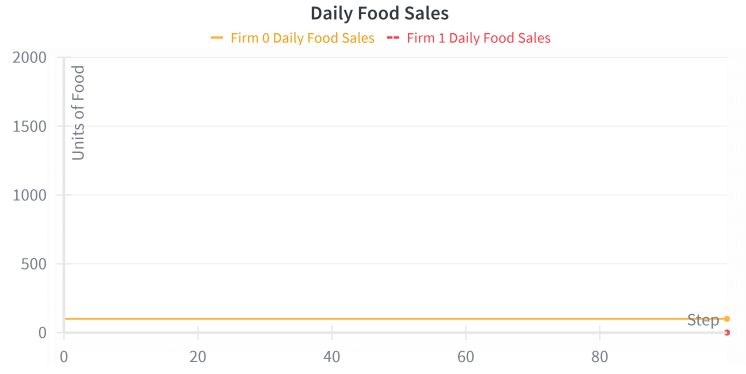

Figure 2: Trial 4 daily food sales.

Figure 3: Trial 4 daily food prices.

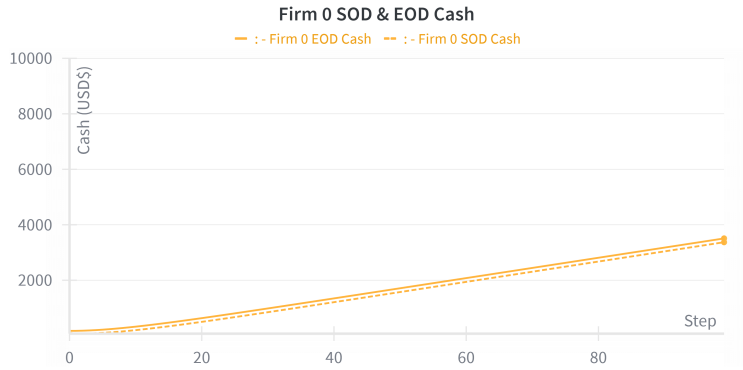

Figure 4: Trial 4 firm 0 cash.

In Trial 4, Firm 0 initially posts a slightly lower price than Firm 1, and therefore wins all of the sales in the first timestep (see Figure3 and Figure 2). This is consistent with the Race to the Bottom hypothesis in which firms posting lower prices win more demand. The unbound complete demand capture is an artifact of the current implementation. Firm 1 is unable to purchase additional inventory for the next timestep but attempts to lower its price:

```
2025-12-04 13:55:03 - INFO - LLM OUTPUT RECURSE 0 {"thought":"I
have 150 units of food inventory and $0 cash.  The previous
price of $2.00 resulted in no sales.  To generate cash and
maximize profit, I need to set a competitive price that encourages
sales while still being profitable. I will try a slightly lower
price than before to hopefully stimulate demand and generate
revenue.","price_food":1.75}
```

Firm 0 maintains its price of $1.75 and again wins all of the sales. Firm 1 becomes insolvent due to lack of revenue and exits the market. This reinforces the hypothesis that the Race to the Bottom scenario leads to monopolization. Although Firm 1 failed, the LLM identified a feasible strategy by lowering the price. With no visibility into other agents, it was forced to guess a competitive price.

With Firm 0 now a monopolist, it fails to exploit the market efficiently. It solely increases price at a constant rate rather than searching for a profit-maximizing price and quantity. Prices rise over 400% while sales volume collapses. This demonstrates a danger of agentic

economies in the Race to the Bottom scenario: naive strategies can dangerously reduce welfare and restrict access to essential goods.

Trial 4 confirms the price war and monopoly formation aspects of the Race to the Bottom hypothesis while also highlighting the potential dangers of inefficiency and decreased welfare.

Trial 3

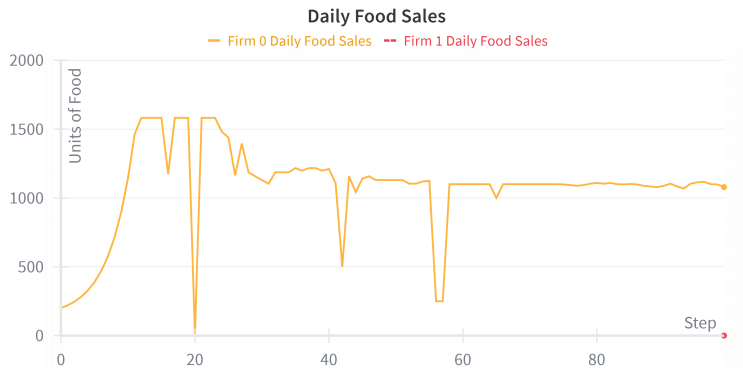

Figure 5: Trial 3 daily food sales.

Figure 6: Trial 3 daily food prices.

In Trial 3, Firm 0 again undercuts Firm 1 early on, winning the price war, seen in Figure 6. Firm 0 maintains consistent pricing aside from a 33% increase from $1.50 to $2.00 around timestep 25 (excluding outliers around timestep 20). Despite this, its profits steadily increase, far surpassing the batch mean. Daily food sales in Figure 5 reveal that the model aggressively increased supply to meet high demand, capitalizing on volume-driven profits.

sf Figure 8 indicates that FirmAgent's consistently found strategies to be profitable after winning the price war.

This behavior aligns with economically sound monopoly theory and demonstrates that the FirmAgent is capable of identifying and executing welfare-improving strategies under the same market rules. The drastic difference in outcome between Trial 3 and Trial 4 indicates that the Race to the Bottom hypothesis correctly predicts monopoly formation. However, these trials indicate that the long-horizon welfare outcomes depend on individual agent reasoning and reflection.

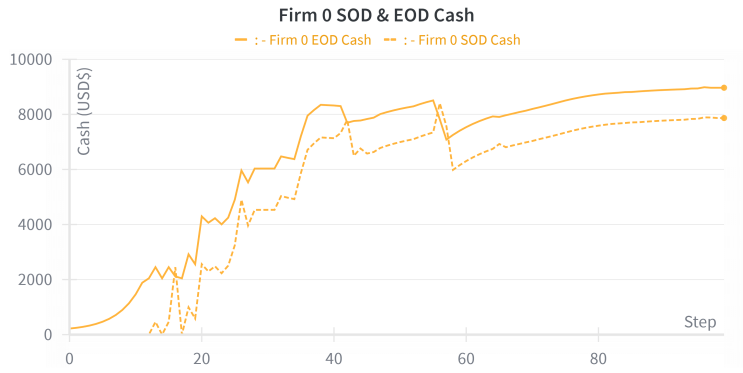

Figure 7: Trial 3 firm 0 cash.

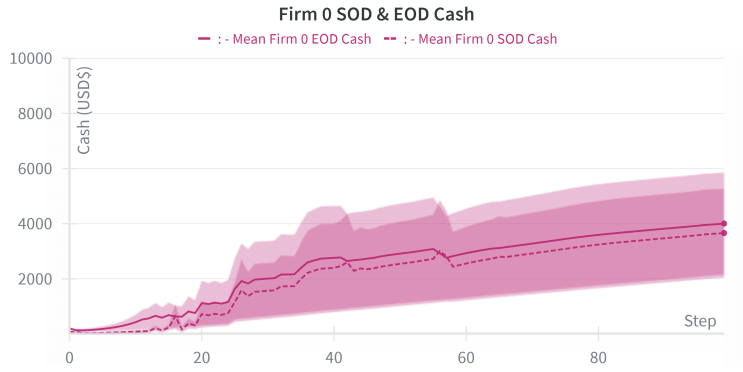

Figure 8: Mean end-of-day (EOD) and start-of-day (SOD) cash for firm 0 across 5 runs plotted with standard error.

6 Discussion

Even in this first batch of trial runs, we see largely rational behavior that one could expect from a human placed in a similar scenario. These reasonable results indicate that the project's current structure is capable of effectively simulating agentic economies for research purposes.

More concerning is the behavior observed in the monopolistic case. In Trial 4, Firm 0 naively increases its price rather than searching for an optimal strategy. Although this maintained daily profits, it depressed sales volume and reduced overall welfare. In contrast, Trial 3 shows a firm aggressively scaling production to match demand, suggesting that the FirmAgent implementation is capable of rational behavior. However, the lack of convergence indicates limitations in the current setup.

The Race to the Bottom scenario arises partly due to missing guardrails that would prevent unrealistic high-frequency trading effects. Potential solutions include introducing rate limits on buy/sell actions, restricting quote polling, and limiting consumer access to firm-level information. Overall, the preliminary results—paired with existing literature—underscore the need for test benches like Agent Bazaar to study agentic markets before such systems appear in real markets.

7 Remaining Work

The simulation still requires significant development before results can inform policy discussions. First, agents need more robust reflection. Trials show firms falling into suboptimal behaviors such as mechanically increasing prices rather than exploring more profitable strategies. Improving reflection should yield more realistic economic behavior.

Aside from the additional experiments required to run the other two scenarios, we do not yet see the mathematical objectives outlined in Section 3 being achieved for the Race to the Bottom (price war) scenario. Revisions to the market clearing operation, as well as the simulations initialization parameters can help yield more interesting results.

Next, future experiments will involve brain swapping across different LLMs to understand how model choice affects market dynamics. Implementing limits on quote polling and action rates will help simulate bounded rationality and prevent unrealistic high-frequency behaviors. Additionally, enabling agents to act as both firms and consumers will support more dynamic economic environments.

Once the simulation is stable and reliable, we aim to build a visualizer that allows users to adjust parameters and observe agents interacting in a spatial environment with statistical overlays. This will make the project more accessible, interpretable, and impactful.

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A Source Code

The source code for *The Grand Agent Bazaar* is available via a public GitHub repository. The repository contains the full simulation framework and agent implementations: https://github.com/CameronCrow/AIBazaar